SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Ethical Hacking **COURSE CODE: ITA14**

COURSE OUTCOME COVERED

CO2: Foot printing Tools: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to

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Question 1: OSINT Email Enumeration Assessment using theHarvester

Objective

To gather publicly available information related to tesla.com using theHarvester and analyze how attackers could use the information during reconnaissance.

Tool Used

- Kali Linux
- theHarvester

Commands Executed

theHarvester -d tesla.com -b crtsh

theHarvester -d tesla.com -b yahoo

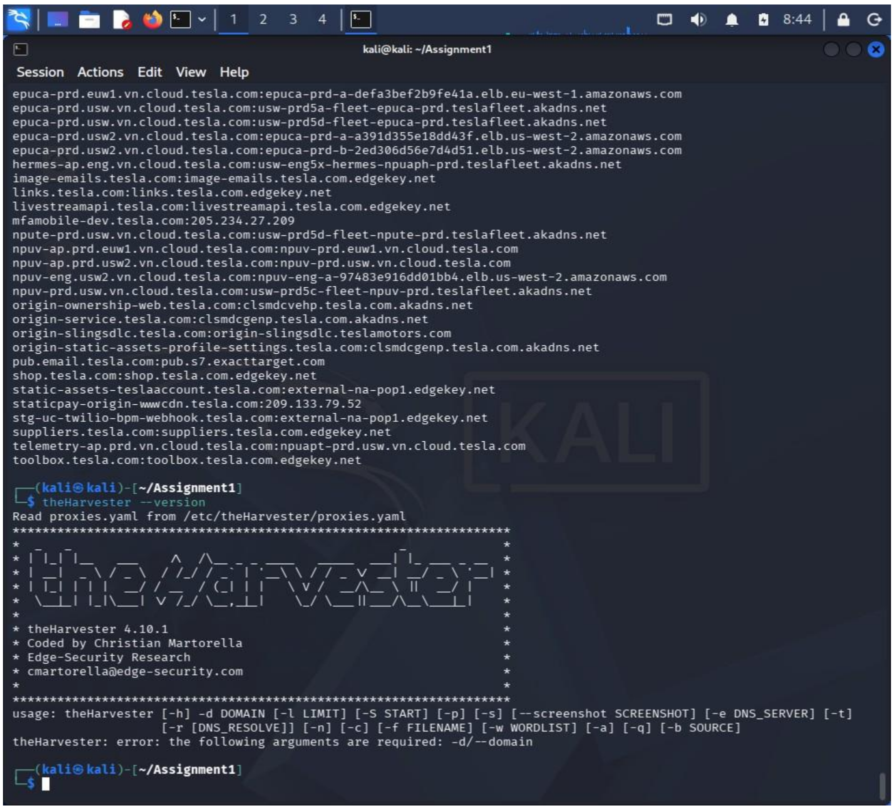
theHarvester -d tesla.com -b subdomaincenter

(Use the commands you actually executed.)

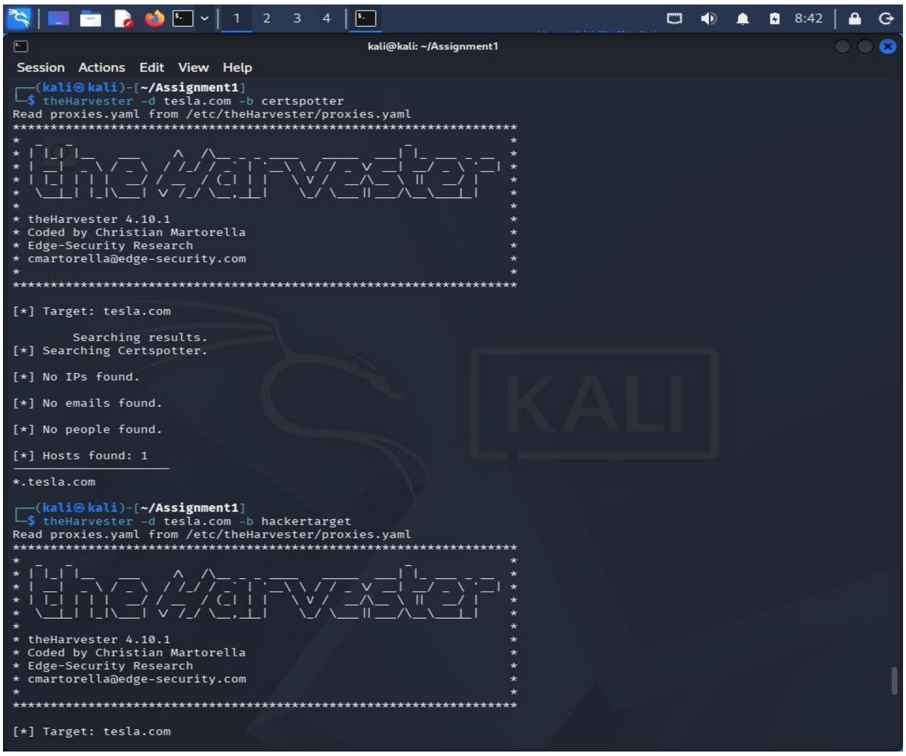
Screenshots

Insert screenshots of:

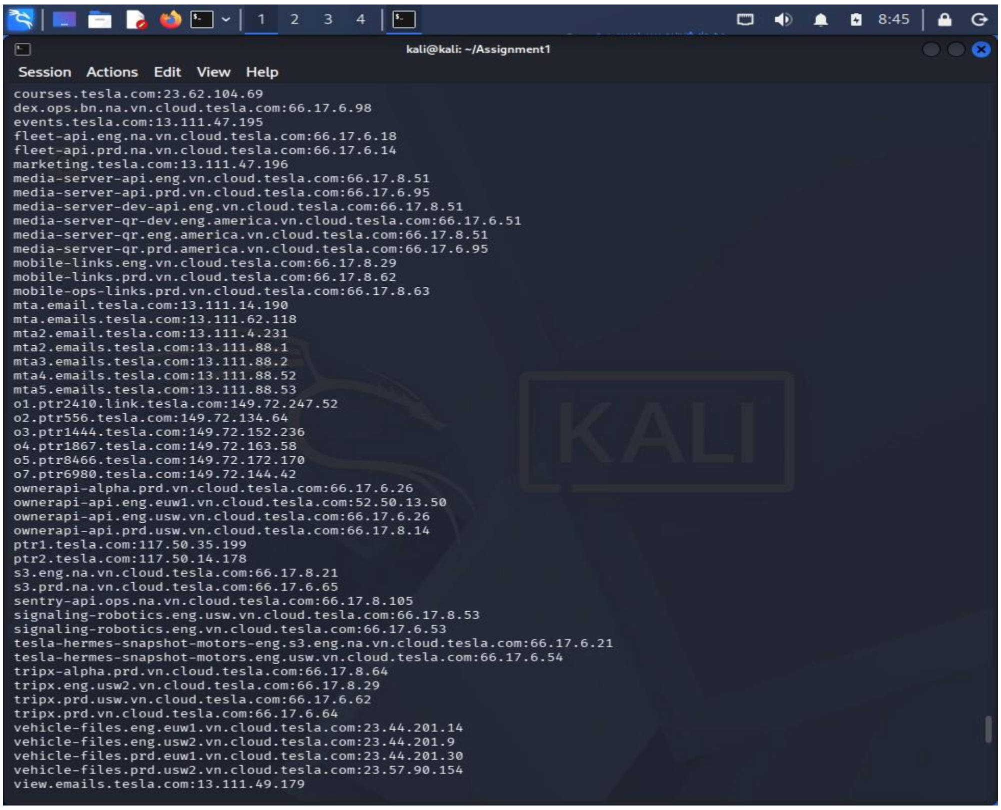
- theHarvester execution



- Subdomains discovered



- Hosts/IPs discovered



Findings

Subdomains

Create a table using the results you obtained.

Subdomain	**IP Address**
mta.email.tesla.com	13.x.x.x
view.email.tesla.com	13.x.x.x
ptr1.tesla.com	117.x.x.x

(Add more from your results.)

Email Addresses

If no emails were found:

No public email addresses were discovered during enumeration.

Hosts and IP Addresses

Include the hostnames and IPs you collected.

Categorization of Email Roles

Example:

Role	Example Email
Support	support@company.com
Sales	sales@company.com
HR	hr@company.com
Administration	admin@company.com
Technical Team	tech@company.com

If none were found, write:

No email addresses were discovered; therefore role categorization could not be performed.

Risk Analysis

Spear-Phishing

Attackers can use discovered domains and hostnames to create convincing phishing emails.

Credential Harvesting

Fake login portals can be created using discovered organizational information.

Impersonation Attacks

Attackers may impersonate internal departments.

Countermeasures

1. Limit public exposure of employee email addresses.
2. Use privacy protection for DNS records.
3. Conduct security awareness training.
4. Implement SPF, DKIM, and DMARC.
5. Use Multi-Factor Authentication (MFA).

6. Monitor information leakage.

Conclusion

OSINT can reveal valuable organizational information that attackers can leverage during reconnaissance.

Question 2: Phishing Awareness Simulation using GoPhish

Objective

To simulate a phishing campaign and measure user awareness.

Tool Used

- GoPhish
- Kali Linux

Campaign Setup

User Group

Dummy Employees

Name	Email
John Doe	john@testlab.local
Jane Smith	jane@testlab.local
David Kumar	david@testlab.local
Sarah Lee	sarah@testlab.local

Email Template

Subject:

Action Required: Password Reset

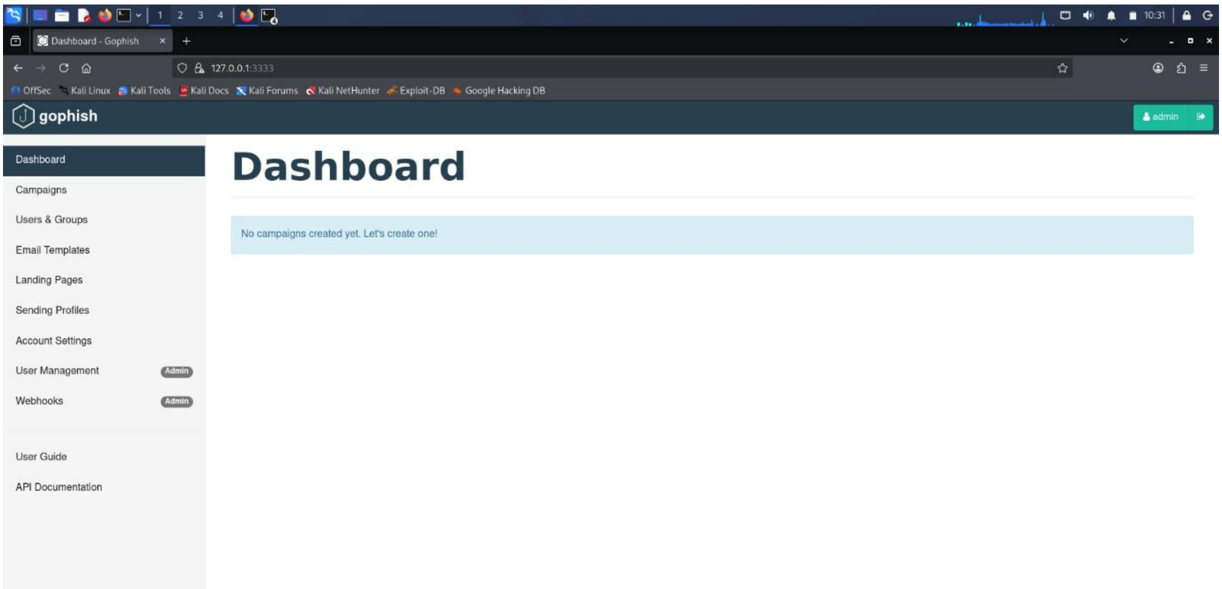
Landing Page

Password Reset Portal

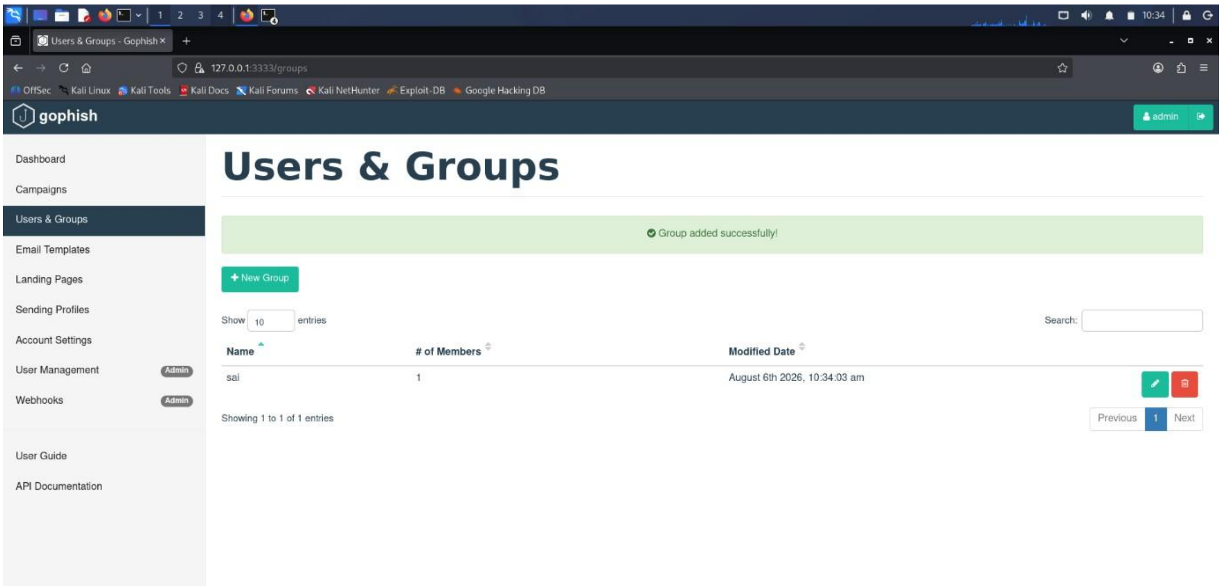
Screenshots

Insert screenshots of:

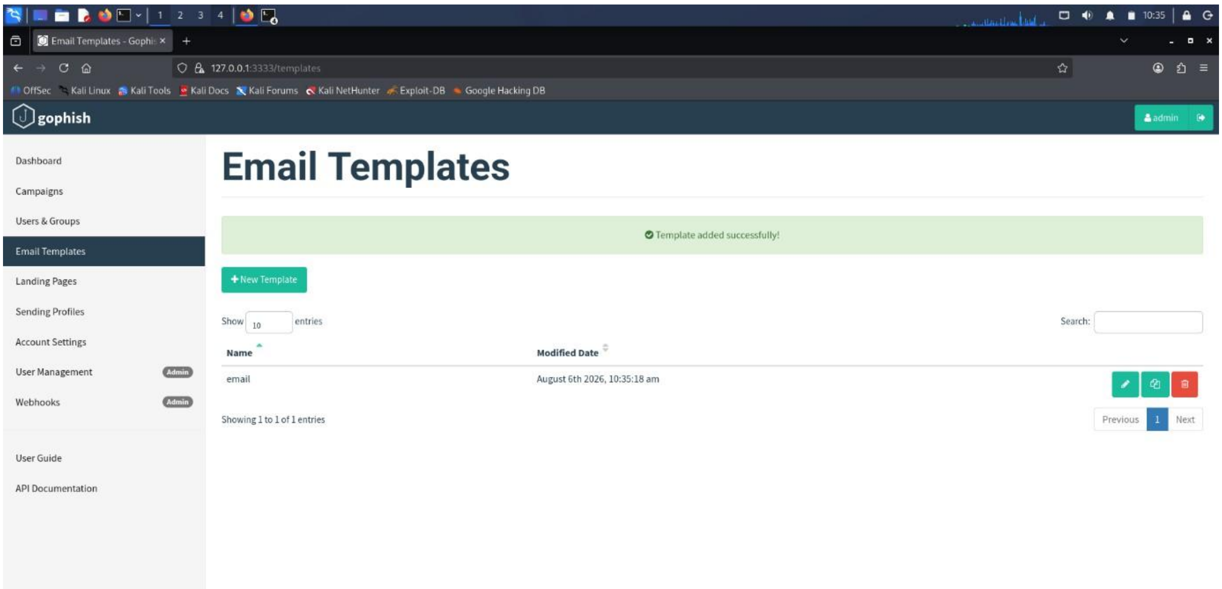
1. GoPhish Dashboard



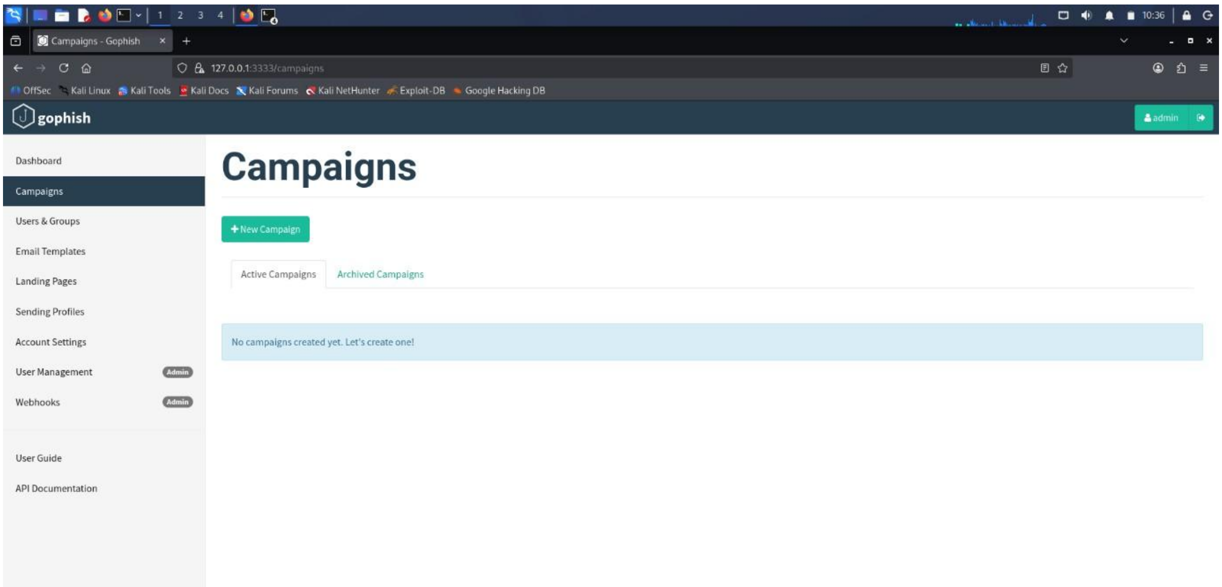
2. User Group Creation



3. Email Template Creation



4. Landing Page Creation



Campaign Results

Use the values shown in your dashboard.

Campaign Results Table

Metric	Value
Total Emails Sent	**4**
Successfully Delivered	**4**
Emails Opened	**3**
Links Clicked	**2**
Credentials Submitted	**1**

Delivery Rate

Formula:

(Delivered / Sent) × 100

Open Rate

(Opened / Sent) × 100

Click Rate

(Clicked / Sent) × 100

Recommendations

1. Conduct regular phishing awareness training.
2. Verify sender identities.
3. Enable MFA.
4. Avoid clicking unknown links.
5. Report suspicious emails immediately.
6. Deploy email security controls.

Conclusion

The phishing simulation highlighted user susceptibility to phishing attacks and the need for ongoing awareness programs.

Question 3: Nmap Aggressive Scan Assessment

Objective

To gather information about scanme.nmap.org using Nmap Aggressive Scan and analyze social engineering risks.

Tool Used

- Nmap
- Kali Linux

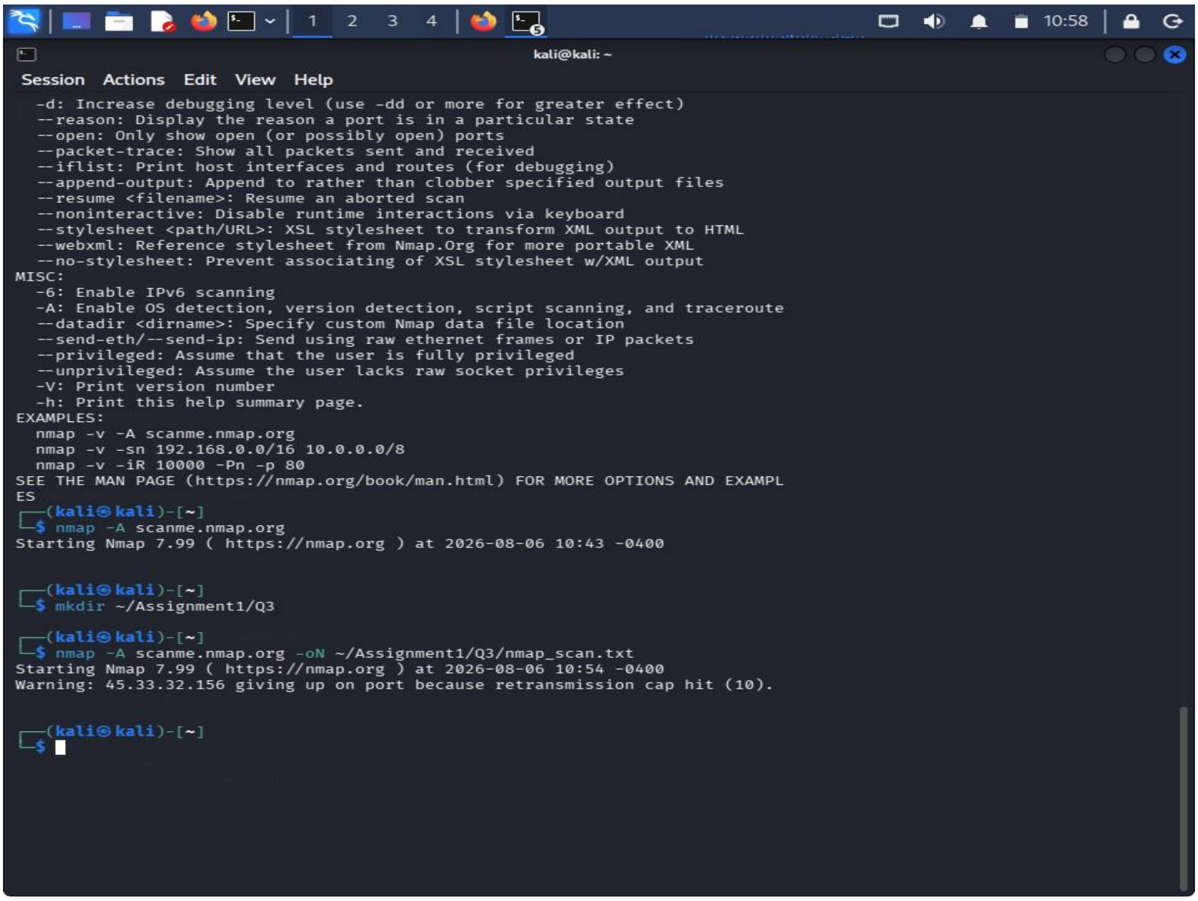
Command Executed

nmap -A scanme.nmap.org

Findings

Open Ports

Port	Service
22	SSH
80	HTTP
9929	NPING Echo
31337	Elite



Service Versions

List detected versions from your scan.

Operating System

Linux-based operating system (or whatever Nmap reported).

Attack Methodology

Reconnaissance

Gather service and OS information.

Social Engineering

Create phishing emails pretending to be IT Support.

Example Subject:

Critical SSH Service Upgrade Required

Objective

Steal administrator credentials through a fake login page.

Defensive Recommendations

1. Minimize service banner exposure.
2. Apply security patches regularly.
3. Enable MFA.
4. Restrict administrative access.
5. Conduct awareness training.
6. Monitor suspicious activity.

Conclusion

Nmap revealed system information that could help attackers craft targeted social engineering attacks.